

AURA

Automated Unified Returns & Resolutions Architecture

Technical Document

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1 System Overview

AURA is a multi-role complaint and returns resolution platform designed to handle the full lifecycle of customer grievances — from submission and AI-assisted verification to merchant response and final resolution. The system connects three primary stakeholders: customers, merchants, and support teams, with AI intelligence woven throughout the evidence pipeline.

1.1 Core Design Philosophy

AURA was built on a key insight: *the quality of a complaint resolution is only as good as the quality of evidence collected*. Traditional image-based complaint systems provide insufficient context for complex cases — they cannot capture product condition across angles, record customer explanations, or authenticate evidence reliably.

This led to AURA’s defining architectural decision: a **video-based verification pipeline** that transforms complaint handling from static evidence collection into live, AI-assisted, multimodal assessment.

1.2 Role-Based Interface Architecture

AURA exposes three distinct interface layers, each optimised for its user type:

- **Customer Interface** — complaint initiation, video session, and resolution tracking
- **Merchant Workspace** — per-customer return/complaint management and analytics
- **Support Team Interface** — ticket queue, workload analytics, and resolution actions

2 System Architecture

2.1 Architecture Diagram

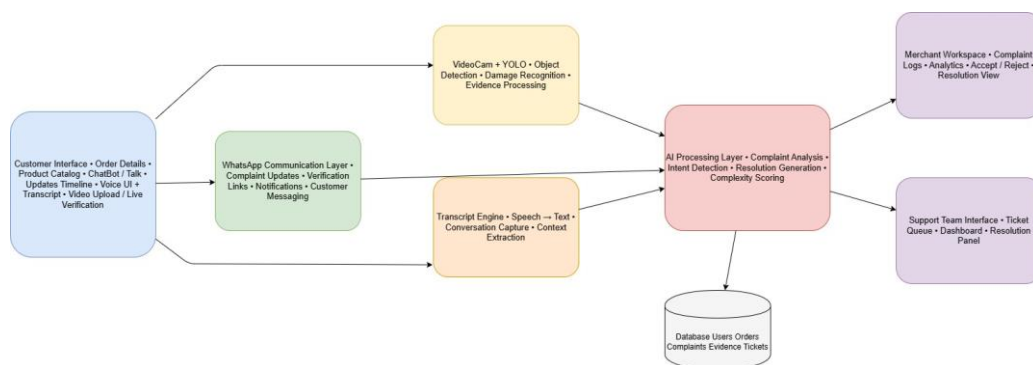


Figure 1: AURA System Architecture — Component & Data Flow Overview

The diagram above illustrates the high-level component layout and data flow across all three interface zones.

2.2 Component Breakdown

2.2.1 Customer Interface

- **Order Details & Product Catalog** — browses current orders and product listings

- **Updates Timeline** — real-time tracking of complaint and delivery statuses
- **Talk & ChatBot** — conversational entry points to initiate a complaint or get help
- **Conversational AI Dashboard** — hosts the live video verification session
- **Voice UI + Transcript** — real-time speech transcription during the session
- **VideoCam-YOLO** — computer vision module that processes the live video feed

2.2.2 Merchant Workspace

- **Customer-wise Return/Complaint Logging** — per-customer complaint registry
- **Data Analytics** — aggregated metrics on return rates, categories, and resolution times
- **Acceptance Button** — one-click acceptance/rejection of resolution proposals

2.2.3 Support Team Interface

- **Ticket Queue (Ticket XXX)** — prioritised list of open cases
- **Charts Dashboard** — visual analytics for workload and resolution tracking
- **Resolution Bar per Entry** — contextual action panel for each ticket

2.3 Data Flow

The core processing pipeline operates as follows:

1. Customer initiates complaint via ChatBot or Talk interface
2. System schedules a live video verification session
3. During the session, **VideoCam-YOLO** processes the live feed in real time, detecting product condition, damage indicators, and authenticity signals
4. Audio is transcribed concurrently, generating a structured transcript
5. Analysed data (YOLO output + transcript intelligence) is forwarded to the AI agent for complaint processing
6. The agent generates a resolution proposal which is surfaced to both the Support Team (via the ticket queue) and the Merchant (via the workspace acceptance button)
7. Resolution status propagates back to the Customer's Updates Timeline

3 Key Implementation Decisions

3.1 Shift from Image-Based to Video-Based Evidence

Initial design used static image submission for complaint verification — a straightforward approach that mirrors most existing returns systems. During development, this was identified as a critical weakness:

- Images lack *temporal context* — they cannot show how damage occurred

- Single-angle images miss important condition signals
- Static media is easier to manipulate or misrepresent
- Images cannot capture the customer’s explanation or intent

The redesign shifted to live video verification sessions. This change was not cosmetic — it restructured the entire evidence pipeline. The result is a significantly richer evidence set: spatial product inspection, live customer narration, AI-detected visual signals, and timestamped proof of session authenticity.

3.2 AI vs. Deterministic Code Boundary

AI / LLM Handles	Deterministic Code Handles
YOLO-based damage/product detection	Ticket creation, assignment, and status updates
Resolution proposal generation	API calls to Shopify for order data
Complaint complexity scoring	Routing logic based on ticket priority rules
Natural language responses in Chat-Bot	Data persistence and audit logging

Table 1: AI vs. Deterministic Code boundary in AURA

The line was drawn at **predictability**. Any operation that needs to be auditable, reversible, or guaranteed to execute in fixed time uses deterministic code. AI is used where the value comes from language understanding or visual recognition — tasks where rules-based systems would either fail or require prohibitive engineering effort.

3.3 Multimodal Fusion Design

YOLO-based visual processing and transcript analysis run as **parallel pipelines** during the live session. Neither alone is sufficient:

- YOLO without transcript context may flag cosmetic wear as damage
- Transcript without visual verification is susceptible to unsubstantiated claims

The fusion layer reconciles signals from both modalities before generating the resolution proposal, reducing both false positives and false negatives.

4 Failure Handling & Graceful Degradation

4.1 Shopify API Failures

- Order data fetches are wrapped in retry logic (exponential backoff, 3 attempts)

- If Shopify is unreachable, the complaint flow continues with user-submitted order details only — a visible warning is shown in the merchant workspace indicating data may be unverified
- Ticket creation does not block on Shopify data; it falls back to a manual data entry mode

4.2 LLM / AI Response Failures

- All LLM calls are wrapped in try/catch with a single retry on malformed or empty responses
- If the retry also fails, the system falls back to a pre-defined response template and flags the ticket for human review
- YOLO inference failures default to *'inconclusive'* visual assessment — the ticket is escalated rather than auto-resolved
- The system never presents a fabricated AI response to the user; it always prefers an honest fallback over a confident wrong answer

4.3 Unexpected User Input

- ChatBot inputs are sanitised and length-capped before LLM submission
- Structured fields (order IDs, claim types) are validated client-side and re-validated server-side
- Off-topic or adversarial inputs to the chatbot return a polite redirection prompt rather than an error state

4.4 System Degradation Model

AURA is designed to **degrade gracefully** rather than fail completely. The priority order is:

1. Full AI-assisted video verification *(nominal path)*
2. Asynchronous video upload with delayed AI processing
3. Image-based submission with manual support team review
4. Text-only complaint submission with full human handling

Each level below the nominal path increases human involvement but ensures the customer can always submit a complaint and receive a response.

5 Known Limitations & Future Improvements

5.1 Current Limitations

- YOLO model is trained on general product categories — accuracy degrades on niche or highly specialised products

- Transcript accuracy drops for non-English speakers or strong regional accents — multilingual ASR is not yet integrated
- The merchant acceptance interface is intentionally simple; bulk resolution workflows are not yet supported
- There is no persistent memory across chat sessions — each complaint conversation starts fresh

5.2 What We Would Improve With More Time

- Train a domain-specific YOLO model on e-commerce product categories (electronics, apparel, packaged goods) for higher detection accuracy
- Integrate multilingual ASR (e.g., Whisper large-v3) to support regional language speakers
- Add a merchant analytics dashboard with trend analysis, return rate forecasting, and category-level insights
- Build asynchronous agent workflows so complex cases can be processed offline without blocking the customer session
- Implement memory/context persistence so repeat customers receive continuity across complaints
- Add confidence scoring UI — show the support agent how confident the AI is on each verification signal so human review is appropriately calibrated

6 Design Evolution: From Images to Video

AURA was initially designed around image-based evidence submission for complaint verification because images are straightforward to collect and process. However, during development it became clear that this approach closely resembled many existing complaint and claim systems and did not provide enough contextual information for more complex customer cases.

Static images often fail to capture important signals such as:

- Product condition from multiple angles
- Customer explanations and intent
- Temporal context (when and how damage occurred)
- Authenticity indicators that distinguish genuine from staged evidence

To create a stronger and more differentiated solution, the design was re-evaluated and shifted toward a video-based verification pipeline. This led to the introduction of live verification sessions, transcript generation, conversational interactions, and multimodal evidence analysis using visual processing (YOLO) combined with transcript intelligence.

Moving from images to video significantly improved evidence richness and gave AURA a more intelligent verification process while distinguishing it from traditional complaint-handling systems.

Design Insight:

This redesign taught us that sometimes improving a system is not about optimising the current approach, but rethinking the entire evidence pipeline around better sources of information.